

Discrimination and Male–Female Earnings Differentials

ECON 300 - Labour Economics

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Learning Objectives

- Describe the factors giving rise to male–female pay gaps and the role of discrimination.
- Outline methodologies used to estimate discriminatory pay gaps and their challenges.
- Explain how discriminatory gaps can persist even with competitive pressures.
- Discuss pros and cons of policy initiatives to combat discrimination.
- Explain the evolution of the gender pay gap and the mechanics of pay equity.

Interpretation

This chapter separates **observed wage differences** from **true discrimination**. Not every pay gap is discriminatory; some may reflect differences in experience, occupation, hours, or job interruptions.

Main Questions

- How can equally productive men and women be paid different wages in competitive markets?
- Will competitive forces dissipate discrimination?
- How do we measure discrimination credibly in the data?
- How much discrimination exists in Canada? What about other groups?
- Which policies have been adopted, and which are effective?

Example

If a man and a woman with the same schooling, experience, and job duties are paid differently, economists ask whether the difference comes from unmeasured productivity, job choice, or discrimination.

What is Labour-Market Discrimination?

Definition

Labour-market discrimination is unequal treatment of groups with the same productivity, or the same productivity-related traits.

- **Employers** may discriminate through wages, hiring, assignments, or promotions.
- **Co-workers** may discriminate because of prejudice, misinformation, or job-security concerns.
- **Customers** may have discriminatory preferences that firms respond to.

Key Distinction

A **pay gap** is not automatically discrimination. A gap becomes discrimination when equally productive workers are treated differently because of group identity.

- Demand theories
- Supply theories
- Noncompetitive theories
- Queuing and efficiency-wage considerations

Interpretation

These theories explain why wage gaps may arise, why they may persist, and why market competition alone may not fully eliminate them.

Demand Theories of Discrimination

- Lower demand for female labour implies lower wages and employment.
- Possible channels:
 - underestimation of female productivity
 - statistical discrimination
 - signalling stories

Definition: Statistical Discrimination

Employers use group averages as a shortcut when they cannot perfectly observe individual productivity.

Example

If employers incorrectly believe women are more likely to leave the labour force, they may offer lower wages even to women who plan to stay continuously employed.

Interpretation of Demand-Side Discrimination

- If employers perceive a group as less productive, the labour demand curve for that group shifts left.
- This reduces both equilibrium wages and employment.
- Even if the beliefs are false, the wage gap can persist if employers act on them.

Economic Intuition

Demand-side discrimination operates like a reduction in the perceived marginal revenue product of the targeted group.

Supply Theories of Discrimination

- Increased supply of female labour can depress wages if productivity or matching suffers.
- **Crowding hypothesis:** segregation into “female-type” jobs creates excess labour supply in those occupations, lowering wages.
- **Dual labour market:** men may be concentrated in core jobs and women in secondary-sector jobs with lower wages.

Definition: Crowding

Crowding occurs when barriers restrict one group to a narrow set of jobs, increasing labour supply there and pushing wages down.

Examples of Supply-Side Mechanisms

- If women are discouraged from entering engineering, more women may end up in clerical or care-sector jobs, increasing labour supply there.
- This can lower wages in female-dominated occupations even when productivity is not lower.
- Occupational segregation may therefore create a wage gap without direct employer prejudice in a single firm.

Example

If nursing and elementary teaching become heavily female-dominated while skilled trades remain male-dominated, relative wages may diverge because labour is unevenly distributed across occupations.

Noncompetitive & Queuing Theories

- **Noncompetitive theories:** adjustment costs and imperfect information can sustain wage gaps.
- **Queuing / Efficiency-wage theories:** firms may pay above-market wages to reduce turnover, increase morale, or raise effort.
- Queues for good jobs can interact with gender differences and reinforce unequal outcomes.

Interpretation

When wages are not fully flexible or information is imperfect, discrimination can persist longer than simple competitive models suggest.

Why Competition May Not Eliminate Discrimination

- In a simple Becker-style model, discriminatory firms should lose profits and eventually shrink.
- In practice, labour markets are not perfectly competitive.
- Search frictions, mobility costs, imperfect information, and institutional barriers slow adjustment.

Key Point

Competitive pressure may reduce discrimination, but it does not guarantee its disappearance.

Productivity Differences: Choice or Discrimination?

- Differences in acquired human capital may reflect:
 - shorter expected labour-force attachment
 - higher turnover or absenteeism
 - household and caregiving responsibilities

Interpretation

Observed wage gaps may partly reflect differences in experience or job continuity, but these differences themselves may be shaped by social norms and constraints.

Choice, Constraints, and Discrimination

- Some differences are often presented as “choices,” but choices are made within constraints.
- Caregiving responsibilities, lack of childcare, and social expectations can affect experience and occupational choice.
- Therefore, the boundary between “choice” and discrimination is often blurry.

Example

If women take more career interruptions because childcare is costly or unavailable, the resulting wage gap is partly tied to institutions, not just preferences.

Setup: Oaxaca(-Blinder) Decomposition

Let

$$\ln W = X\beta + \varepsilon$$

be estimated separately for men and women.

Then the mean log wage gap can be decomposed as:

$$\ln W_m - \ln W_f = (\bar{X}_m - \bar{X}_f)' \hat{\beta}_m + \bar{X}_f' (\hat{\beta}_m - \hat{\beta}_f)$$

- First term: part **explained** by differences in characteristics.
- Second term: **coefficient effect** or unexplained gap, often interpreted as discrimination.
- The decomposition depends on the chosen reference wage structure.

Definition and Interpretation of Oaxaca Decomposition

Definition

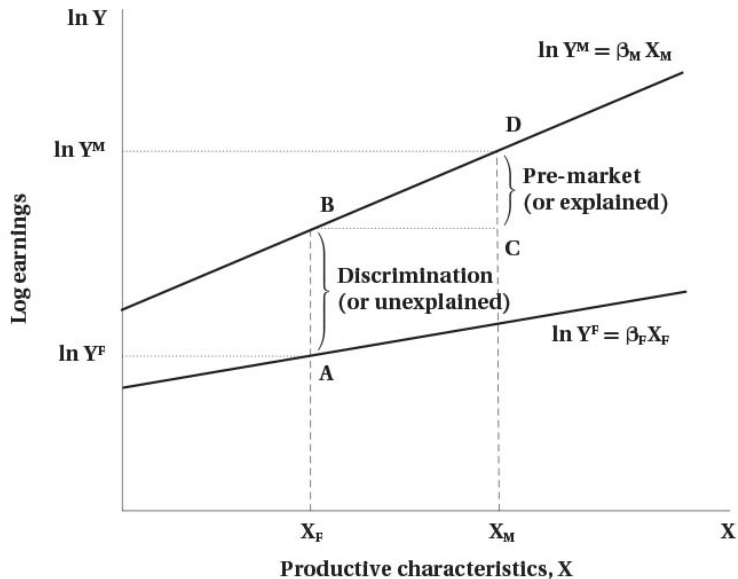
The **Oaxaca-Blinder decomposition** is a method for splitting an average wage gap into:

- a part due to differences in measured characteristics, and
- a residual part due to differences in returns to those characteristics.

Interpretation

The unexplained component is not automatically pure discrimination, but it is often used as a benchmark estimate of discriminatory treatment.

Figure 12.1: Graphical Illustration of the Oaxaca Decomposition



Reading Figure 12.1

- The vertical axis is log earnings; the horizontal axis is productive characteristics X .
- One wage equation is estimated for men and another for women.
- The total wage gap can be split into:
 - an **explained** or pre-market part from different characteristics,
 - an **unexplained** part from different coefficients.

Key Idea

If men and women had the same productive characteristics, any remaining gap is interpreted as evidence of unequal returns.

Worked Example: Determinants of Earnings

Estimated earnings equations:

$$\ln W_m = 0.40 + 0.07S + 0.06EXP + 0.05Married + 0.01Children$$

$$\ln W_f = 0.01 + 0.10S + 0.04EXP - 0.02Married - 0.05Children$$

Group means:

$$\bar{X}_m = (S = 14, EXP = 10, M = 0.5, C = 0.6), \quad \bar{X}_f = (S = 15, EXP = 7, M = 0.5, C = 0.6)$$

Goal

Use these equations to compute the raw gap and counterfactual outcomes.

Worked Example: Predicted Values and Raw Gap

Predicted male log wage:

$$E(\ln W_m) = 0.40 + 0.07(14) + 0.06(10) + 0.05(0.5) + 0.01(0.6) = 2.01$$

Predicted female log wage:

$$E(\ln W_f) = 0.01 + 0.10(15) + 0.04(7) - 0.02(0.5) - 0.05(0.6) = 1.75$$

Raw female-to-male ratio:

$$\frac{E(\ln W_f)}{E(\ln W_m)} = \frac{1.75}{2.01} \approx 0.87$$

Interpretation

Women in this example are predicted to earn about 87% of men's predicted log earnings.

Worked Example: Counterfactual Pay Structures

Female characteristics evaluated at male coefficients:

$$\ln W_f^* = 0.40 + 0.07(15) + 0.06(7) + 0.05(0.5) + 0.01(0.6) = 1.90$$

Male characteristics evaluated at female coefficients:

$$\ln W_m^* = 0.01 + 0.10(14) + 0.04(10) - 0.02(0.5) - 0.05(0.6) = 1.77$$

Implications:

$$\ln W_f^* - E(\ln W_m) = 1.90 - 2.01 = -0.11$$

$$E(\ln W_f) - \ln W_m^* = 1.75 - 1.77 = -0.02$$

Interpretation

Counterfactuals help answer: *What would women earn if they were paid according to the male pay structure?*

Additional Evidence on Male–Female Pay Differentials

- Studies examine wage, sex, institutional, and household discrimination channels.
- Average female earnings are often reported below male earnings in some datasets and periods.
- Key question: What would women earn if they were treated like men while holding productive traits fixed?

Empirical Evidence for Other Groups

TABLE 12.1

Ethnic Wage Differentials, Various Groups Born in Canada or United States

	Black	South Asian ^a	Southeast Asian ^b	Chinese	Aboriginal
Canada					
Total gap	0.293	0.536	-0.022	0.193	0.179
"Discrimination"	0.171	0.222	0.023	0.058	0.091
United States					
Total gap	0.328	0.176	-0.079	-0.205	0.304
"Discrimination"	0.123	0.055	-0.007	-0.047	0.136

NOTES: The total gap is the percentage by which the earnings of whites exceed those of the ethnic group, reflecting both the differences in their wage-determining characteristics and the differences in pay for the same characteristics. The latter is the differences in the regression coefficients from an Oaxaca-type decomposition as discussed in the text, and is labelled here as "Discrimination."

^a E.g., Bangladesh, India, Pakistan

^b E.g., Philippines, Korea, Vietnam

How to Read Table 12.1

- The **total gap** measures the overall difference in wages relative to the reference group.
- The **discrimination** column reports the unexplained component after controlling for measured characteristics.
- The gap can differ across countries and groups because institutions, labour markets, and histories differ.

Interpretation

A positive unexplained gap suggests lower pay than expected given observed characteristics.

- **Conventional equal pay:** same job, same establishment.
- **Equal value / Pay equity / Comparable worth:** compare predominantly male and female jobs of equal value.
- **Equal employment opportunity (EEO):** protect against discrimination in recruiting, hiring, and promotion.
- **Affirmative action / Employment equity:** targeted plans for designated groups.
- **Facilitating policies:** childcare and family-friendly practices.

Equal Pay and Pay Equity

- Equal pay legislation requires equal pay for equal value within the same establishment.
- Job evaluation must be free of gender bias.
- All Canadian jurisdictions except Alberta have some relevant legislation.

Interpretation

Equal pay is narrower than pay equity. Equal pay compares similar jobs; pay equity can compare different jobs judged to be of equal value.

Comparable Worth: Design Features

- Define gender predominance.
- Choose a job-evaluation procedure.
- Specify adjustment procedures and possible exemptions.
- Enforcement and complaint mechanisms matter in practice.

Example

A hospital may compare compensation for a male-dominated technical occupation with a female-dominated care occupation and adjust wages if the jobs are evaluated as equal in value.

EEO and Employment Equity

- **EEO** can raise female employment and wages by reducing barriers in recruiting, hiring, and promotion.
- **Employment equity** in federal jurisdiction applies to designated groups including:
 - women,
 - visible minorities,
 - persons with disabilities,
 - Aboriginal peoples.

Interpretation

These policies aim not only to equalize wages but also to improve access to better jobs.

Facilitating Policies

- Childcare support
- Parental leave
- Flexible scheduling
- Family-friendly workplace practices

Interpretation

These policies do not directly set wages, but they can reduce structural constraints that generate wage gaps.

Example

Affordable childcare may increase continuous labour-force attachment and reduce experience gaps between men and women.

- Labour-market discrimination is unequal treatment of equally productive workers.
- Demand, supply, noncompetitive, and queuing theories provide different explanations for wage gaps.
- Oaxaca-type decompositions separate explained from unexplained wage differences.
- Evidence shows persistent gender and group differentials.
- Policy tools include equal pay, pay equity, EEO, employment equity, and facilitating policies.

Thank You!